Feature Reduction of SVM-Based Classifier For Predictive Maintenance OF Rotating Machines

Ahmed Mokhtar ElSayed
School of Electrical and Electronic Engineering
Coventry University
Ahmed.Raafat@tkh.edu.eg

2025 7th Novel Intelligent and Leading Emerging Sciences Conference (NILES) | 979-8-3315-9168-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/NILES68063.2025.11232435

***Abstract*—This study proposes a predictive maintenance approach aimed at diagnosing motor faults, particularly bearing failures, at an early stage to avoid costly unplanned machine downtime. The proposed method leverages vibration analysis and a Support Vector Machine (SVM) classifier to identify fault types based on frequency-domain features. A key contribution is the optimization of both model accuracy and computational efficiency through dimensionality reduction. The ReliefF feature selection algorithm was applied to identify the most significant statistical features which reduced the number of features from 11 to 7 while maintaining a classification accuracy of 93.3%. As a result, the reduced model exhibits a 20.8% improvement in prediction speed, which makes it better suited for edge-computing deployment.**

Keywords— Industry 4.0, Fault Diagnosis, Predictive maintenance, Vibration Analysis, Machine Learning, DSP.

I. Introduction

Bearings-problems are very common cause for machines-failure [1]. They resemble about 40% of the major problems that face induction machines [2]. This has been the main drive for selecting the bearings as the major problem to be predicted. To achieve this predictive diagnosis approach, the root causes of bearings failure should be captured. The root causes will be diagnosed and captured through vibration analysis and machine-learning deployment. Furthermore, the machine learning model will be optimized through a feature-dimensionality reduction technique to fasten the speed of the prediction as an early stage for developing edge-computing capabilities for the system. This paper does not only aim to achieve a great accuracy in the fault diagnosis while considering a single axis but also to study the significance of the most common used statistical features in this field in industry 4.0 to filter out those with the least importance. This is expected to enhance the generalization of the machine-learning model and increase its speed of prediction. The studied vibrational dataset is obtained from [3]. This vibrational dataset consists of 5 lab-scale water pumps with different physical conditions. One motor is running at completely normal conditions. The second and the third motors encounter different magnitudes of unbalances; 6 gm.cm and 27 gm.cm respectively. The fourth motor deliberately experienced a misalignment offset of 3mm from the center. The fifth motor's bearings are damaged (inner and outer raceway faults). The signal-processing techniques and feature-extraction methods applied to the raw data in this study are consistent with state-of-the-art approaches reported in the literature.

II. Literature Review

Numerous studies have been conducted in this field, highlighting multiple possible routes toward achieving a robust solution, ranging from the type of data collected to the choice of signal-processing techniques and the application of suitable machine learning models. Starting with the data-collection technique, some papers install acoustic sensors (e.g. Microphones) to collect sound data of the operating machine [4]. This approach depends on the variation of the sound data when there is an abnormality in the machine. Other approaches that rely upon accelerometers for anomaly detection, had also been introduced. Accelerometers give insights of the vibrations of the rotating parts of the machine in order to detect any abnormality within its activity. To make it clear, depending on the power and magnitude of the vibrations, the activity is classified as normal or abnormal. This approach is used by [5] and [6]. A third approach has also been studied in the past couple of years. This approach focused on the electrical signal analysis in detecting the abnormality of the machine. As stated by [2], this method depends on installing current and voltage sensors to analyze the electrical signals coming out of the machines. When it comes to the signal processing steps, there are three different signal-processing techniques that could be done; Time-domain analysis, frequency-domain analysis and time-frequency analysis which is also known as wavelet transform. [4].Numerous papers had previously compared between analyzing the raw data through these three domains. Wavelet transforms are used in a recognizable amounts of papers such as [1] which used a type of wavelet transform called daubechies transform, yet declared that the frequency-domain analysis is the most common used technique in detecting bearings-failure. Another type of wavelet transformed had been utilized by [7] called DWT for detecting shaft misalignment under variable loading conditions. On the other hand, several papers applied Fast-Fourier transform as a signal-processing technique, proofing that it is the most common technique and proofing its great accuracy of detecting different patterns of data. [6] Used FFT for rotating-shafts unbalance detection. Moreover FFT is used by the paper whose dataset was used [3] and used also by [7] besides utilizing the DWT technique. Afterwards, in order to close the main cycle of this project, the most suitable type of machine-learning model should be chosen in order to focus on the features selection later as a crucial step in this study. The research published by [8] applied CNN (Alex Net architecture) for detecting the levels of wear that encountered the different parts of the bearings and achieved an accuracy of 98.84% and the performance of the SVM in the same study reached up to 98.5 %. ANN also was utilized by [1] to detect different types of bearings' faults reaching an accuracy of 94.7 %. Not to mention that a crucial review paper that is published by [9] which compared and summarized the pros and cons of the most relevant models in this field. The comparison is summarized in the following table.

979-8-3315-9168-7/25/$31.00 ©2025 IEEE
Authorized licensed use limited to: FUNDACAO UNIVERSIDADE DO RIO GRANDE. Downloaded on June 26,2026 at 15:38:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I Models performances

Algorithm	Advantage	Disadvantage
SVM	High accuracy	Slow response for larger data
K-NN	Robust to noise	Slow in real time
ANN	Parallel processing	Hardware and over fitting
DT and RF	Easy to understand	Over fitting
Naïve Bayes	Robust to noise	Takes Time for Training
Deep Learning	Low computation cost and extracts its own features	Large data and time for training

Finally, a very effective SVM had been implemented by [5]. When it comes to feature engineering, there are common features that are used in almost every paper for training the machine-learning models. It is also worth mentioning that the majority of the studies take into consideration the vibrations of the three axis (x, y and z) while doing the signal processing and extracting the features. For example, 18 features were used in [7] composing of 6 various statistical features utilized in the three different axis. The features are skewness, kurtosis, minimum, maximum, RMS and entropy. Moreover, 7 statistical features had been used in [1] for bearing-fault detection using ANN which are kurtosis, skewness, variance, standard deviation, speed and load. Another published review paper in this field declared the common statistical features which are mean, standard deviation, skewness and kurtosis [9]. Also [3] utilized FFT and then extracted statistical features from the output of the FFT in the three axis (x, y and z) as well. The features obtained from each axis are: shape factor, RMS, Impulse Factor, peak-to-peak, kurtosis, crest factor, mean, standard deviation and skewness (27 concatenated features).

III. Methodology

The whole process could be summarized via the following block diagram.

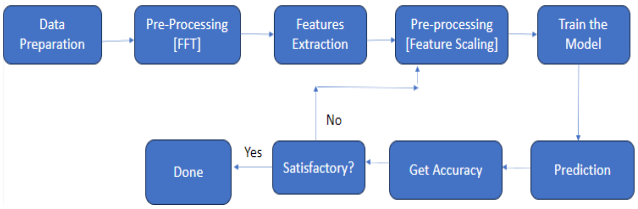


Fig. 1 Process Brief

A. Type of data and faults of interest

Starting with the choice of the fault-detection technique, vibration analysis using accelerometers will be used in this project. This is because vibration analysis, compared to sound analysis, is considered to be a more effective way for detecting a motor's fault. Vibration analysis is more sensitive to mechanical problems and is the most common used technique. It carries out dynamic information and shows immediate response to any change as stated by many papers [1]. This study targets the early prediction of the bearing's failure through capturing two of its main root causes which will eventually lead to the bearings failure itself. It is stated in [10] that some of the root causes for bearings are shaft misalignment and unbalance. This had been supported by a real industrial problem discussed in the same study where the vibrations of a 75-hp blower motor indicated a reduced efficiency. This observed behavior reflected a shaft-misalignment problem.

B. Preprocessing technique

Despite being applicable, it is still challenging to extract important features of signals from the time-domain analysis only. Hence there are two most common used methods for analyzing the raw signal; frequency-domain analysis and wavelet transform analysis. It is worth mentioning that the wavelet transform is the most accurate and latest DSP technique. Moreover it is applied in a variety of projects as stated in the previous section. However the chosen one in this project would be the frequency domain (FFT specifically). This technique would be applied on the time-domain raw data. FFT is chosen because it is easier to interpret and it is proofed to be as efficient as required. Moreover, to ensure a fair comparison with [3], the same preprocessing techniques were applied, thereby isolating dimensionality reduction as the main point of difference. This technique is also applied in various experiments as well, and proofed to be a good technique for features' extraction as deduced in the literature review. Furthermore, when it comes to the comparison between the time domain and the frequency domain analysis, it would be more advantageous to use the frequency-domain analysis (FFT) because frequency domain gives the luxury of breaking down the vibrations into individual components and thus the sources of defects can be detected separately [11].

C. Dataset Utilized

The vibrational dataset that will be used is prepared by [3]. The data is composed of 4000 csv files. Each file contains around 100,000 data points (vibrations x, y, z and timestamp) because data is acquired at a sampling rate of 20 KHz for 5s. The FFT of the 4000 files is computed, resulting in complex outputs. Afterwards, the magnitudes of the FFT of the raw data points (**of the x-axis only**) in each of the 4000 files are extracted and saved. The next stage included applying the features-extraction algorithm so that each csv file is replaced with a single multi-dimensional data point (11 features) so that the resultant output is 4000 different data points stored in a single file available for training as follows:

- File 1:1000 data points – Normal condition
- File 2:1000 data points - Misalignment condition
- File 3: 1000 data points – Unbalance features
- File 4: 1000 data points- Damaged-bearings features

These files are all combined in a single CSV file showing 4000 data points with their corresponding target (machine condition).

D. Feature Engineering and machine learning model selection

1) Features

According to numerous papers in this field, the most commonly used statistical features in training models are: ***Mean, Minimum, Maximum, Standard deviation RMS, Crest factor, Peak-to-peak, Skewness, Kurtosis,***

Authorized licensed use limited to: FUNDACAO UNIVERSIDADE DO RIO GRANDE. Downloaded on June 26,2026 at 15:38:08 UTC from IEEE Xplore. Restrictions apply.

Shape Factor *and* ***Impulse factor.*** Note that these features are the same used by [3].

2) Model selection

Even though, the CNN is widely used, it will be an excluded option because this study has a potential to have edge-computing capabilities. Adding a computer-vision task will consume a lot of processing power. Besides CNN, ANN was a strong competitor that is implemented in a variety of papers. However, since it depends on neural networks as well, the SVM will be advantageous since it is expected to consume less power than neural networks. By referring to table I, a direct comparison has been made between ANN and other machine learning algorithms. The table proofs that ANN has hardware limitations compared to others. Not to mention, the over fitting disadvantage which is not required at all in this project (the project requires a generalized machine learning model that is adaptable to different types of machines). Briefly, the selected machine learning model is the SVM. By referring to table 1 again, SVM shows a great accuracy. Moreover, as previously mentioned, SVM had been implemented in various papers and showed great-accuracy results. Even though, some papers mentioned other algorithms and techniques of machine learning that could be used, they compare the performance of those proposed algorithms with SVM classifier most of the time as being the classical machine learning algorithm.

IV. Results

For the classical vibration analysis without the intervention of the machine learning, each fault could be matched with a unique signature onto its FFT plot; this is actually one of the advantages of using vibration analysis as previously mentioned since each rotating component will appear onto the spectrum correspondent to its frequency of rotation. With that being said, the dataset is tested for verification through plotting a random file from each type of fault to make sure each fault matches its anticipated signatures.

A. FFT plots of data (Analysis by inspection).

It is worth mentioning that the fundamental frequency of the machine is 50 Hz since its RPM is 3000.

The FFT of the x-axis vibrations of a two randomly chosen data files (100,000 data points each); one for misalignment and one for unbalance, is calculated and plotted as in the following figures (Fig. 2-5).

By observing the following plots, in case of an unbalance, the dominant frequency component is almost at 50 Hz (see fig. 2). This could be validated in the normalized plot (fig. 4) where the x-axis values are divided by 60 so that they are resembled as multiples of the fundamental frequency. Thus in fig. 4, the dominant component was almost at 1x (The fundamental frequency). On the other hand, by inspecting the normalized misalignment plot (fig. 5), it could be declared that the dominant frequency component is at 2x. These observations match the theory of the classical vibration analysis when studying the misalignment and unbalance signatures (2x and 1x respectively). [12]

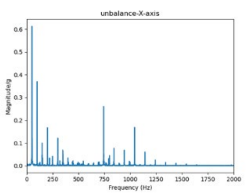


Fig. 2 Unbalance spectrum signature

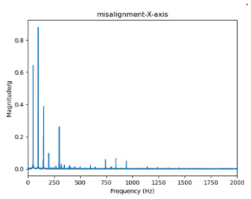


Fig. 3 Misalignment spectrum signature

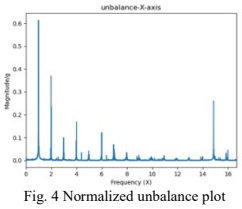


Fig. 4 Normalized unbalance plot

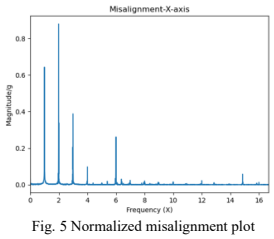


Fig. 5 Normalized misalignment plot

B. Accuracy of the model, before and after feature scaling

First of all, grid search was initially applied to compare kernel types, where the RBF kernel showed superior performance. For subsequent experiments, the RBF kernel was employed with default hyperparameters (C = 1.0, γ = 'scale'), as further tuning seemed unnecessary given the stable and satisfactory results obtained. The model's accuracy was evaluated on unseen data using a train–test split of 70%–30%, ensuring that 30% of the input data remained unseen during testing the accuracy. The first run of the trained SVM showed a poor performance of 37.1%. By comparing the range of values of each feature to the other, it is concluded that there are some features with huge magnitudes compared to others which could introduce a bias to the model. Therefore, the proposed solution is to standardize the features using the following formula:

$$\text{Standardization}=\frac{(X-x_mean)}{standard\ deviation}.$$

Fortunately, after scaling the features, the performance of the model had been enhanced, reaching an accuracy of 93.3%. The following figure shows the confusion matrix of this high-accuracy trained model.

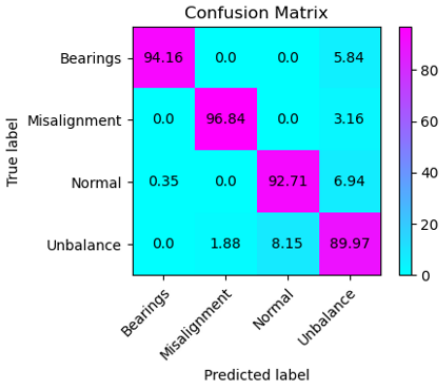


Fig. 6 Confusion Matrix

Some of the parameters that the confusion matrix show are the bearings recall of the model = 94.2%, misalignment recall of the model= 96.8%, normal-condition recall of the model= 92.7%, unbalance recall of the model= 89.97%

Authorized licensed use limited to: FUNDACAO UNIVERSIDADE DO RIO GRANDE. Downloaded on June 26,2026 at 15:38:08 UTC from IEEE Xplore. Restrictions apply.

C. Feature Ranking Algorithm for dimensionality reduction

As one of the aims of this study is to maintain the high performance of the model with the least possible number of features to reduce overfitting, achieve generalization and reduce prediction time, two main techniques are implemented; the first technique is extracting features from data of the x-axis only and exclude the y-axis and the z axis data. The second technique involves applying a feature ranking algorithm called ReliefF algorithm so that all the statistical features that are utilized in the majority of the papers will be reduced according to their significance as indicated by the ranking scores. Those ranking scores will be provided by the ReliefF algorithm. This algorithm was proposed previously by [7]. Briefly, ReliefF estimates feature importance by examining each sample and its closest neighbors so that features that keep samples of the same class close while separating those from different classes are assigned higher weights, whereas less informative features receive lower weights. Features score rankings were as follows:

TABLE II Features Ranking

Rank	Feature Name	Importance Score
1	Shape Factor	0.3912
2	RMS	0.3083
3	Standard deviation	0.3066
4	Mean	0.2728
5	Peak to peak	0.2402
6	Maximum	0.2402
7	Skewness	0.0561
8	Crest Factor	0.0333
9	Impulse Factor	0.0333
10	Kurtosis	0.0301
11	Minimum	0.0000

According to table II, only the first seven features will be selected to retrain the model and analyze its performance when the dimension of the model's hyperplane reduced from (4000, 11) to (4000, 7). To validate the new model's performance, some of the data was prepared separately and fed to the model to predict their nature.

TABLE III Test Results

Model Type	SVM	SVM
# of Features	11	7
Time Elapsed/ms	24	19
Accuracy	93.3 %	93.3 %

The table shows that the model's prediction was much faster (down from 24 ms to 19 ms) which is a recognizable change regarding edge-implementations.

V. Conclusion

This study focuses on diagnosing motor faults, particularly bearing failures, which account for about 40% of motor issues, through predictive maintenance. It targets early detection by identifying root causes like shaft misalignment and unbalance, using vibration signals analyzed by a Support Vector Machine (SVM) model. The SVM model achieved a 93.3% accuracy in classifying water pump conditions using 11 common features. To improve generalization and prediction speed, a two-stage dimensionality reduction strategy was applied: using only x-axis vibration data and selecting top-ranked features via the ReliefF algorithm. This approach retained high accuracy while significantly reducing the number of features compared to [3]. Compared to the benchmark study [3], which used the same dataset and classifier with 27 features, this study reduces the input to just 7 features by utilizing single-axis vibration data (instead of triaxial vibrational data), and applying the ReliefF algorithm to take into account the most important features only. Despite a slight reduction in accuracy, the model maintained a high 93.3% classification performance while achieving a 20.8% improvement in prediction speed (reducing latency from 24 ms to 19 ms). This time reduction, though seemingly small, is significant in industrial settings where large volumes of sensor data must be processed in real time. This demonstrates a key contribution: developing a lightweight, efficient SVM-based diagnostic model suitable for real-time, edge-deployable predictive maintenance without sacrificing diagnostic reliability.

VI. Acknowlegement

I would like to express my deepest gratitude to Dr. Khaled ElGeneidy for his exceptional guidance, mentorship, and continuous support throughout this project. His expertise, encouragement, and constructive feedback have been pivotal in its successful completion and in shaping my academic and professional growth. I am also sincerely thankful to Eng. Mohamed Wahba from ElSewedy Electric for generously sharing his expertise in vibration analysis, which provided valuable insights and greatly enriched the technical depth of this work. Finally, I acknowledge Spectro Systems Company for offering the industrial exposure that initially inspired this project.

VII. References

[1] M. Vaibhav, A. Kalhapure, and R. R. Navthar, "Vibration Analysis and Fault Diagnosis of Induction Motor Bearing Using Artificial Neural Network (ANN)," *Int. J. Eng. Tech.*, vol. 2, [Online]. Available: http://www.ijetjournal.org.

[2] E. L. Bonaldi *et al.*, "Predictive Maintenance by Electrical Signature Analysis to Induction Motors," *Induction Mot. - Model. Control*, Nov. 2012, doi: 10.5772/48045.

[3] B. T. Atmaja, H. Ihsannur, Suyanto, and D. Arifianto, "Lab-Scale Vibration Analysis Dataset and Baseline Methods for Machinery Fault Diagnosis with Machine Learning," *J. Vib. Eng. Technol.*, 2023, doi: 10.1007/s42417-023-00959-9.

[4] D. Goyal and B. S. Pabla, "The Vibration Monitoring Methods and Signal Processing Techniques for Structural Health Monitoring: A Review," *Arch. Comput. Methods Eng.*, vol. 23, no. 4, pp. 585–594, 2016, doi: 10.1007/s11831-015-9145-0.

[5] M. Sonowal, B. B. Gogoi, M. Boruah, and J. Kumar Barman, "Health Monitoring of Induction Motor through Vibration Analysis," *ADBU J. Electr. Electron. Eng.* |, vol. 3, no. 1, 2019, Accessed: Jul 09, 2023. [Online]. Available: www.tinyurl.com/ajeee-adbu.

[6] O. Mey, W. Neudeck, A. Schneider, and O. Enge-Rosenblatt, "Machine Learning-Based Unbalance Detection of a Rotating Shaft Using Vibration Data," *IEEE Int. Conf. Emerg. Technol. Fact. Autom. ETFA*, vol. 2020-September, pp. 1610–1617, 2020, doi: 10.1109/ETFA46521.2020.9212000.

[7] A. M. Umbrajkaar, A. Krishnamoorthy, and R. B. Dhumale, "Vibration Analysis of Shaft Misalignment Using Machine Learning Approach under Variable Load Conditions," *Shock Vib.*, vol. 2020, 2020, doi: 10.1155/2020/1650270.

[8] L. A. Pinedo-Sánchez, D. A. Mercado-Ravell, and C. A. Carballo-Monsivais, "Vibration analysis in bearings for failure prevention using CNN," *J. Brazilian Soc. Mech. Sci. Eng.*, vol. 42, no.12, 2020, doi: 10.1007/s40430-020-02711-w.

[9] P. Kumar and A. S. Hati, "Review on Machine Learning Algorithm Based Fault Detection in Induction Motors," *Arch. Comput. Methods Eng.*, vol. 28, no. 3, pp. 1929–1940, 2021, doi: 10.1007/s11831-020-09446-w.

[10] B. Lu, D. B. Durocher and P. Stemper, "Predictive maintenance techniques," in IEEE Industry Applications Magazine, vol. 15, no. 6, pp. 52-60, November-December 2009, doi: 10.1109/MIAS.2009.934444.

[11] Ifm, "Vibration Technology | Frequency Domain - ifm." https://www.ifm.com/us/en/us/real-time-maintenance/vibration/technology/frequency-domain (accessed Jul. 15, 2023).

[12] H. Morton, "Spectrum analysis," *J. Franklin Inst.*, vol. 88, no. 1, pp. 56–62, 1869, doi: 10.1016/s0016-0032(69)90237-3.

Authorized licensed use limited to: FUNDACAO UNIVERSIDADE DO RIO GRANDE. Downloaded on June 26,2026 at 15:38:08 UTC from IEEE Xplore. Restrictions apply.